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Identification of 27 abnormalities from multi-lead ECG signals: an ensembled SE_ResNet framework with Sign Loss function

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Keywords: ECG signal, deep neural network, multi-label classification, ensemble modeling, ECG abnormalities

Abstract

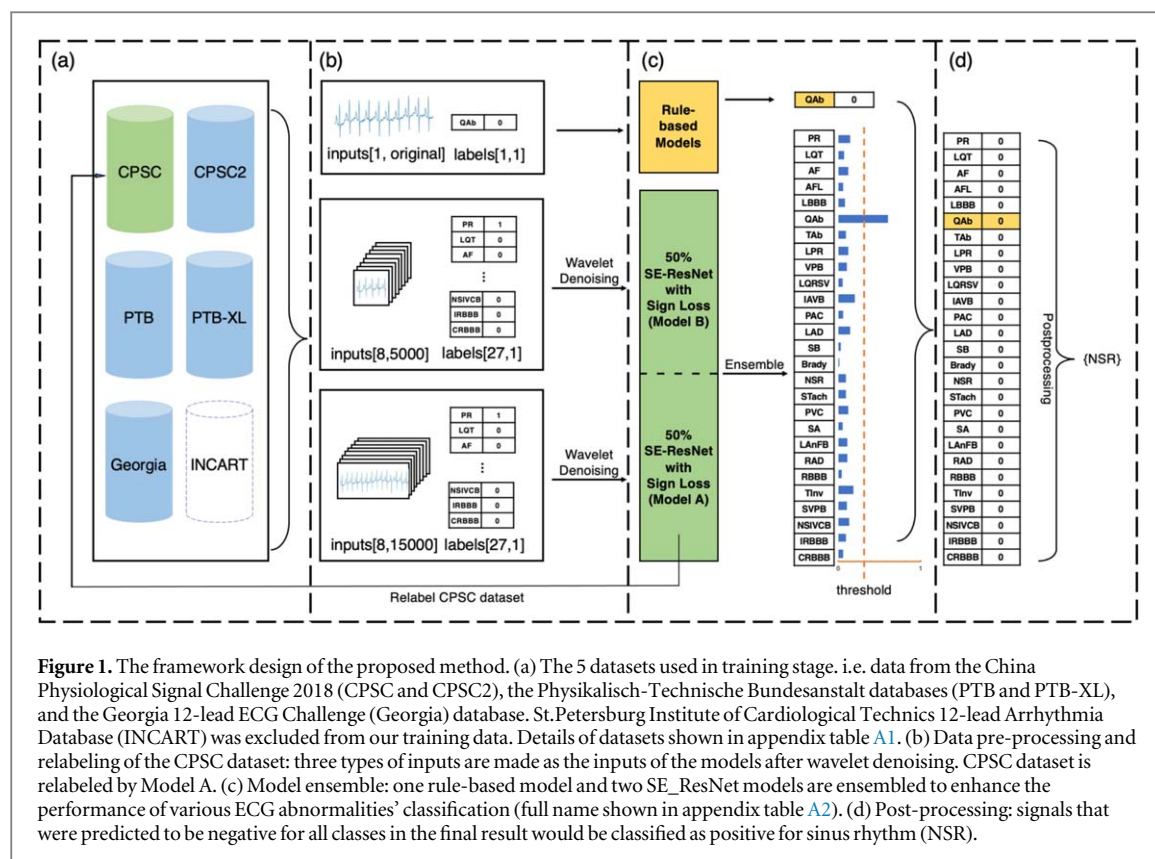
Objective. Cardiovascular disease is a major threat to health and one of the primary causes of death globally. The 12-lead ECG is a cheap and commonly accessible tool to identify cardiac abnormalities. Early and accurate diagnosis will allow early treatment and intervention to prevent severe complications of cardiovascular disease. Our objective is to develop an algorithm that automatically identifies 27 ECG abnormalities from 12-lead ECG databases. **Approach.** Firstly, a series of pre-processing methods were proposed and applied on various data sources in order to mitigate the problem of data divergence. Secondly, we ensembled two SE_ResNet models and one rule-based model to enhance the performance of various ECG abnormalities' classification. Thirdly, we introduce a Sign Loss to tackle the problem of class imbalance, and thus improve the model's generalizability. **Main results.** In the PhysioNet/Computing in Cardiology Challenge (2020), our proposed approach achieved a challenge validation score of 0.682, and a full test score of 0.514, placed us 3rd out of 40 in the official ranking. **Significance.** We proposed an accurate and robust predictive framework that combines deep neural networks and clinical knowledge to automatically classify multiple ECG abnormalities. Our framework is able to identify 27 ECG abnormalities from multi-lead ECG signals regardless of discrepancies in data sources and the imbalance of data labeling. We trained our framework on five datasets and validated it on six datasets from various countries. The outstanding performance demonstrate the effectiveness of our proposed framework.

1. Introduction

Cardiovascular disease is one of the primary causes of death globally, resulting in an estimated 16.7 million deaths each year, according to the World Health Organization Gaziano *et al* (2010). Early and accurate diagnosis of ECG abnormalities can prevent serious complications such as sudden cardiac death and improve treatment outcomes Artis *et al* (1991).

The 12-lead ECG is a cheap, widely available tool to screen for heart disease screening Kligfield *et al* (2007). However, the ECG interpretation requires experienced clinicians to carefully examine and recognize pathological inter-beat and intra-beat patterns. This process is time-consuming and subject to inter-observer variability Bickerton and Pooler (2019). Hence, an accurate algorithm for automated ECG pattern classification is highly desirable.

Some earlier works have reported automated analysis of ECG Martínez *et al* (2004), Minami *et al* (1999), Mahmoodabadi *et al* (2005), Alexakis *et al* (2003). These approaches are mainly based on frequency domain



features, time-frequency analysis, and signal transformations (i.e. wavelet transform and Fourier transform). However, such techniques are not capable of capturing complex features of the ECG signal.

More recently, a number of works have demonstrated the ability of nonlinear machine learning techniques in the field of ECG analysis. Vafaie *et al* (2014) proposed a classifier to predict heart diseases, in which a fuzzy classifier was constructed with a genetic algorithm to improve prediction accuracy. Chen *et al* (2018) designed a gradient boosting algorithm to detect atrial fibrillation. Piece-wise linear splines were used to select features. The performance of those approaches is limited by the choices of input features to the models. The input features are typically carefully selected, which requires time and expert knowledge to verify the feasibility.

In recent years, deep learning and neural networks, especially the convolutional neural networks (CNNs) LeCun *et al* (1995), have gained promising results in many areas such as computer vision Krizhevsky *et al* (2017) and natural language processing Devlin *et al* (2018). CNNs have been used for ECG abnormalities detection based on ECG signal, Xiong *et al* (2018) developed a 21-layer 1D convolutional recurrent neural network to detect atrial fibrillation, which was trained on single-lead ECG data. Sodmann *et al* (2018) proposed a CNN model to improve detection performance by annotating QRS complexes, P waves, T waves, noise, and inter-beat of ECG segments. Elola *et al* (2019) developed two deep neural networks for classifying pulse-generating rhythm and pulse-less electrical activity using short single-lead ECG segments. Warrick and Homsy (2018) designed an ensemble deep learning model for automatic classification of ECG arrhythmias based on single-lead ECGs, which fused the decisions of ten classifiers and outperformed the single deep classifier. Hong *et al* (2019) proposed a multi-level knowledge-guided framework that can extract beat, rhythm, and frequency level features separately. These neural network based solutions considerably improved performances as they are able to better capture the underlying complex and nonlinear relationships in data. Therefore, we also utilize neural networks in our methods.

Most of the previous works focus on one or at most 9 ECG abnormalities' classification Wang *et al* (2019), Luo *et al* (2019), Huang *et al* (2019), Xia *et al* (2019), Elola *et al* (2019), Irusta *et al* (2019), Krasteva *et al* (2020). Therefore, we aim to develop a robust model that can generalize to 27 different types of ECG abnormalities. In addition, most of the existing works only deal with single-lead ECG signals Andreotti *et al* (2017), Billeci *et al* (2017), Bin *et al* (2017), Warrick and Homsy (2018), while in clinical practice, 12-lead ECGs are more commonly used for abnormality detection and diagnosis. Hence, we developed a model that efficiently analyses 12-lead ECG to capture the full picture of potential abnormalities.

The novel contributions of this work can be summarized as follows. *First*, We develop a robust deep learning framework that is able to identify 27 types of ECG abnormalities by analysing eight leads ECG signals from

multiple 12-lead ECG datasets. *Second*, the proposed framework employs a series of novel ECG signal processing methods, including multi-source data pre-processing that mitigate the data divergence problem, model ensemble that improve the framework's generalizability, and Sign Loss function that reduce the negative effects caused by class imbalance. *Third*, to validate the effectiveness of our proposed methods, extensive experiments were conducted on six datasets across the world and examined in the PhysioNet/Computing in Cardiology Challenge (2020). The considerable performance improvement over the state-of-the-art demonstrates the effectiveness of our proposed methods.

2. Methods

In this section, we elaborate our proposed methods in identifying 27 ECG abnormalities from multi-source ECG databases as illustrated in figure 1. We first introduce the datasets we used. Then we present the data pre-processing techniques that reduce the data divergence, the design of the ensemble model that enables efficient multi-label classification with multi-leads ECG, and the Sign Loss in tackling the class imbalance problem. Finally, we present the evaluation metrics and our training setups.

2.1. Datasets

In PhysioNet/Computing in Cardiology Challenge (2020), the challenge data were partitioned into three parts, i.e. training dataset, official validation set, and official test dataset. Organizers made the training data and their corresponding diagnosis publicly available, whereas the official validation dataset and the official test dataset were kept hidden. The official validation dataset helps contestants validate their models. However, the source and size of which have not been published by the organizer. The official test dataset was used to calculate the final score for ranking.

2.1.1. Training dataset

The public challenge training data consists of 43 101 12-lead ECG signals from six different datasets, as summarized in appendix table A1. The sampling frequency of the signals varies from 257 to 1000 Hz, and the length of the signals varies from 6 s to 30 min. There are 111 labeled abnormalities, where multiple labels may correspond to the same type of abnormality. Therefore, 27 types of abnormalities are included in the final scoring metrics. The details of the 27 ECG abnormalities are shown in appendix table A2.

From these data, we created our offline training and validation dataset. Note that the INCART data was excluded from our training data since it has only 74 30 min records with a sampling frequency of 257 Hz and is significantly different from other datasets.

2.1.2. Offline validation dataset

After we processed the original training data as described in section 2.4, we randomly split 80% as the training set (30 172 samples) and 20% as our offline validation set (7544 samples). To further validate our model's generalizability, an external dataset from the Hefei Hi-tech Cup ECG Intelligent Competition TIANCHI—Hefei Hi-tech Cup ECG Intelligent Competition (2020) (Hefei dataset in short) was applied as external validation. Hefei dataset consists of 40000 records of 12-lead ECG signals with a sampling frequency of 500 Hz and length of 10 s. Out of all the records, 6500 records with labels in the 27 types of ECG abnormalities we focused on were randomly selected and formed an external validation set.

2.1.3. Official test dataset

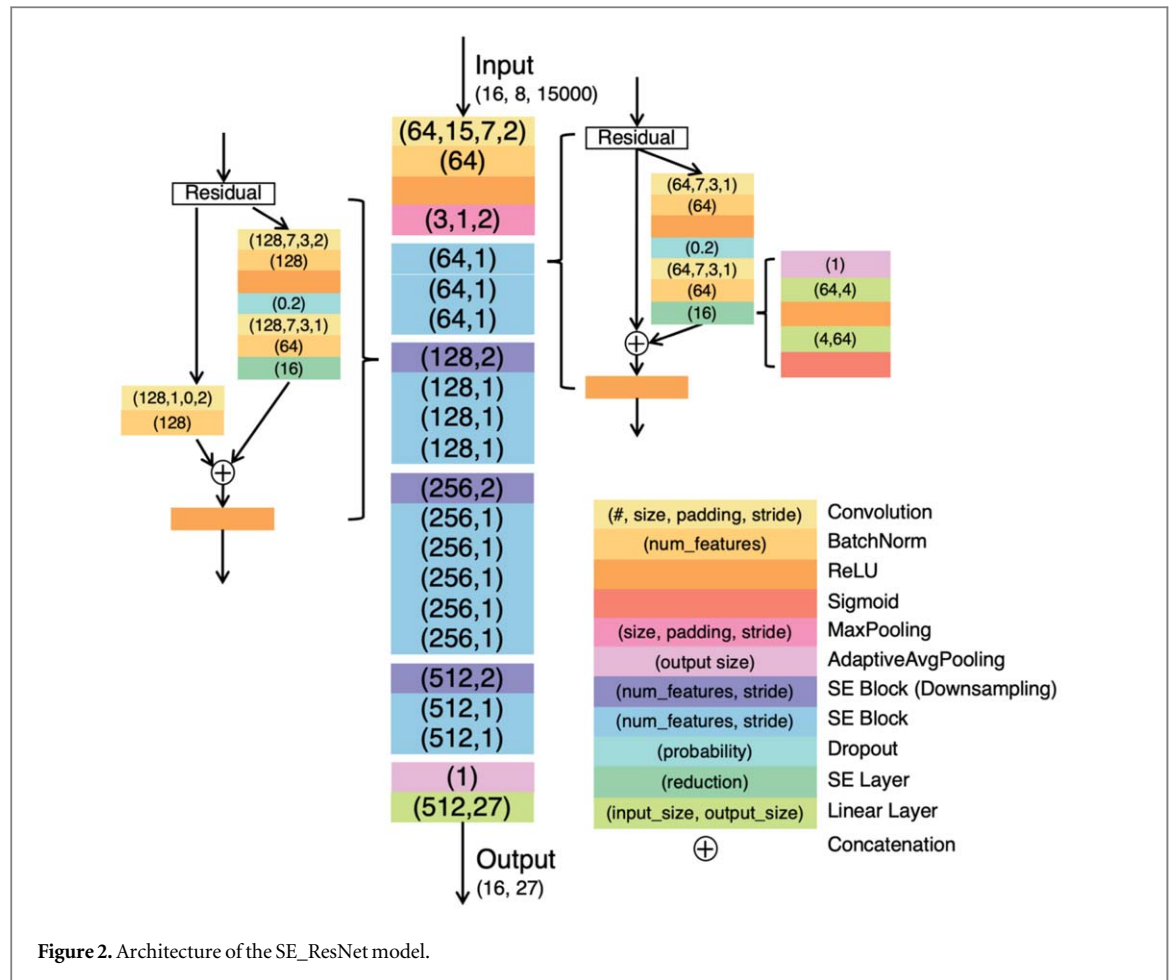
The entire official test dataset contains 11 630 12-lead ECG recordings that were not represented in the training data. The test data was drawn from the following three databases. Test Database 1: A total of 1463 ECGs from the CPSC database. Test Database 2: A total of 5167 ECGs from the Georgia database. Test Database 3: A total of 10000 ECGs from an unspecified US institution.

2.2. Multi-source data pre-processing

The raw data were sampled from different sources, varying in sampling rate, signal amplitude, noise level, etc. To better prepare the data for model training, we adopted the following data pre-processing techniques.

2.2.1. Processing original data

We first exclude data without a label in the 27 scored classes. PTB dataset was downsampled from 1000 to 500 Hz to make the sampling frequency of all training data consistent. Then, we exclude lead III, aVR, aVL, and aVF from the model's input. This is because these four leads are linearly dependent on other leads and can be calculated based on Einthovens Law Kligfield (2002) and Goldberger's equations Goldberger *et al* (2018).



2.2.2. Truncating and padding

For the baseline model (i.e. single SE_ResNet), all input signals were fixed at 30 s in length. This was done by truncating the part exceeding the first 30 s for longer signals and padding the shorter signals with zero. For the other ensembled model, the input length was fixed at 10 s with the same pre-processing method.

2.2.3. Wavelet denoising

Biorthogonal wavelet transformation (bior2.6) was applied to reduce the noise in ECG signals. The numbers of vanishing moments for the decomposition and reconstruction filters were 2 and 6, respectively. The level of refinement was set to be 8, where the high-frequency coefficients in level 1, level 2, and level 8 were set to zero.

2.2.4. Relabeling CPSC data

CPSC dataset was relabeled because the original labels cover only nine classes and the distribution of the labels is significantly different from other datasets. A SE_ResNet was first trained on the original training set and then used for inferring pseudo labels on the CPSC dataset. For each ECG signal in the CPSC dataset, inferred pseudo labels were added as new labels if (1) the inference output probability was higher than 0.8, (2) the labels not in the original nine labels, and (3) the labels were in the 27 officially scored labels.

2.3. Multi-label classification with multi-leads ECG

To improve the accuracy and efficiency of multi-label classification with multi-leads ECG, we adopted the following methods.

2.3.1. SE_ResNet

One important feature of the 12-lead ECG signal is that the information contained differs in different leads due to the difference in signal voltage intensity and amplitude variation. Different ECG abnormalities may be more apparent in specific leads. An equal importance of different leads could cause information losses, leading to misdiagnosis.

Therefore, we use SE_ResNet Hu *et al* (2018) as our primary model to capture the distinctive information in each of the multi-leads ECG signals, and the SE_ResNet architecture is shown in figure 2. More specifically, we

integrate sixteen squeeze-and-excitation (SE) blocks into the ResNet He *et al* (2016) structure. Each SE block applies a squeeze operation and an excitation operation to the input tensor. The squeeze operation compresses the global spatial information by using global average pooling and produces an embedding of the global distribution of feature responses for each channel. Thus all the layers of the network can use the information from the global receptive field. The excitation operation takes the embedding as input and captures the channel-wise dependencies, then produces weights for each channel. Consequently, these weights are applied to previously learned feature maps and realize the feature re-calibration.

In this way, some leads could be given higher weights, leading to a better prediction performance for multi-leads ECG classification. Our baseline model is a SE_ResNet model with an input length of 30 s. Meanwhile, to minimize the effect of padding on the shorter signals, another SE_ResNet model was trained with the input length of 10 s and ensembled with the baseline model.

2.3.2. Rule-based model

The baseline model did not perform well on bradycardia, which indicates the heart rate is slower than one beat per second. In other words, the R–R intervals between consecutive heartbeats are always longer than 1 second. Therefore, we use Pan and Tompkins algorithm Pan and Tompkins (1985) to detect the R-peaks on lead I. Then, R–R intervals could be easily calculated from the positions of consecutive R-peaks. The pseudocode of the rule-based model for bradycardia is shown in algorithm 1.

Algorithm 1. Rule-based bradycardia classifier.

```

Input: List of R-R intervals
Output: Classification of bradycardia
brady_beats = 0;
foreach R-R interval
    if  $1s \leq \text{length of interval} \leq 1.6s$ 
        brady_beats += 1
if brady_beats/the number of R-R intervals  $\geq 0.5$ 
    return True
else
    return False

```

However, a high recall and low precision were observed when we only applied the rule-based model to classify bradycardia. It could be due to the low quality of labels in the datasets. Therefore, the rule-based model was only applied when the prediction of bradycardia is negative from the ensembled model. The pseudocode for the final bradycardia prediction is shown in algorithm 2.

Algorithm 2. Final bradycardia prediction.

```

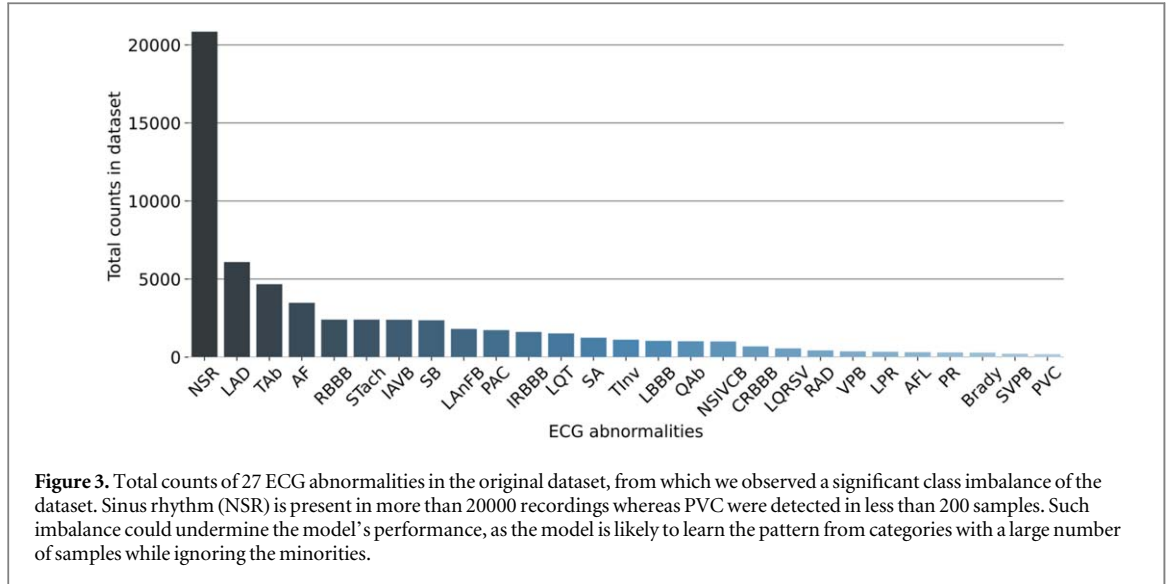
Input: Prediction from ensembled model
        Prediction from rule-based model
Output: Final classification of bradycardia
if Prediction from rule-based model is False
    return False
else
    return Prediction from ensembled model

```

(c) *Model ensemble.* Fusion is a common and effective method to improve generalizability. The idea of fusion in this paper is to combine two models that receive different length input signals. More formally, the ensemble model output g is calculated by:

$$g(x|\theta_1, \theta_2) = 0.5 \cdot f(x_{l_1}|\theta_1) + 0.5 \cdot f(x_{l_2}|\theta_2), \quad (1)$$

Where x_{l_1} and x_{l_2} are the truncated input with length of $l_1 = 10$ s and $l_2 = 30$ s respectively (corresponding to the data length of 5000 and 15000), and $f(x|\theta)$ is the output of each SE_ResNet model with the parameters of θ . Different data lengths could provide the model with multiple views of data. We further applied a NSR post-processing to the model's prediction. Specifically, as the signal contains at least one type of abnormality, signals



that were predicted to be negative for all classes were classified to be positive for sinus rhythm (NSR) due to its dominant ratio in the dataset.

2.4. Class imbalance

2.4.1. Sign loss

A significant issue observed in our data was class imbalance, shown in figure 3, which resulted in predictions biased towards the majority class. Inspired by Sun *et al* (2019), we designed an improved multi-label Sign Loss for our model training. The loss is defined as follows:

$$\text{Sign}(p) = \begin{cases} y - 2py + p^2, & |y - p| < 0.5 \\ 1, & |y - p| \geq 0.5 \end{cases} \quad (2)$$

$$\text{Loss} = \sum_{i=1}^{27} \text{Sign}(p_i) \times \text{BinaryCrossEntropyLoss}(p_i, y_i), \quad (3)$$

where y denotes the ground truth and p denotes the model's estimated probability for $y = 1$. For the correctly classified labels, a coefficient smaller than 1 was multiplied to the default binary cross-entropy loss. By doing so, the accumulated loss from a large number of true negative labels became smaller, and the loss from the misclassified labels became more prominent. Furthermore, the gradient of this loss function has a significant change around 0.5, which enables our models to capture this change. Thus the optimal binarization threshold will also be close to 0.5 and more robust.

2.5. Evaluation metrics

We adopted the official evaluation metrics from PhysioNet/Computing in Cardiology Challenge (2020) and Alday *et al* (2020). To conform to real-world clinical practice, where some misdiagnoses are less harmful than others, the misdiagnoses that end in similar outcomes or treatments as the ground truth diagnoses will still be awarded partial credit. Only 27 anomalies of the total 111 anomalies in 6 datasets were included in the final evaluation.

To be more specific, $C = [c_i]$ defined as the collection of our predictions. The multiclass confusion matrix is $A = [a_{ij}]$, where a_{ij} is the normalized number of recordings in a database that were classified as belonging to class c_i but actually belong to class c_j . The a_{ij} is calculated by $a_{ij} = \sum_{k=1}^n a_{ijk}$, where a_{ijk} is defined by

$$a_{ijk} = \begin{cases} \frac{1}{|x_k \cup y_k|}, & \text{if } c_i \in x_k \text{ and } c_j \in y_k \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

For n ECG signal recordings, x_k is the set of positive labels and y_k is the collection of positive predictions for the k th recording. Hence, the $|x_k \cup y_k|$ is the number of unique classes with a positive label and predictions for recording k . A reward metrics $W = [w_{ij}]$ is defined by the challenge organizer, where w_{ij} denotes the reward for a positive classifier output for class c_i with a positive label c_j . Then, the unnormalized score will be calculated by equation (5)

Table 1. Performance of different models on offline validation datasets. All the models are developed based on our baseline model. From this table we select Model 6 as our best model. N denotes number of leads, L denotes input length, *Tech1* denotes multi-source data pre-processing, *Tech2* denotes rule-based model for bradycardia, *Tech3* denotes Sign Loss and *Tech4* denotes NSR post-processing.

Model	N	L	<i>Tech1</i>	<i>Tech2</i>	<i>Tech3</i>	<i>Tech4</i>	Offline validation score	Hefei validation score
Model 1	12	15000	×	×	×	×	0.682	0.241
Model 2	12	15000	✓	✓	×	×	0.689	0.249
Model 3	12	15000	✓	✓	✓	✓	0.673	0.300
Model 4	8	15000	✓	✓	✓	✓	0.674	0.300
Model 5	8	5000	✓	✓	✓	✓	0.674	0.236
Model 6	8	15000 and 5000	✓	✓	✓	✓	0.683	0.319

$$\text{Unnormalized_Score} = \sum_{i=1}^m \sum_{j=1}^m w_{ij} a_{ij}, \quad (5)$$

$$\text{Normalized_Score} = \frac{\text{Unnormalized_Score} - \text{Inactive_Score}}{\text{Correct_Score} - \text{Inactive_Score}}. \quad (6)$$

After normalization by equation (6), a score of 1 will be assigned to the classifier that always predicts the true label, while a score of 0 will be assigned to an inactive classifier. The Inactive Score is the score for an inactive classifier that always outputs a normal class, while the Correct Score is the score for the model that always predicts the true class. The detailed calculation of Normalized_Score is shown in PhysioNet/Computing in Cardiology Challenge (2020). The Normalized Score will be in the range between 0 and 1, and the higher the score indicates the better performance of the model. We evaluate the Normalized Score in our offline validation set and Hefei validation set.

2.6. Training setup

The proposed model was trained with a batch size of 16 for 19 epochs as the training loss was not further decreasing. The model parameters were optimized with the Adam optimizer Kingma and Ba (2015). During training, the learning rate was set as 0.001 and rescheduled to 0.0001 at the 13th epoch. The optimal binarization threshold was found to be 0.36 on the offline validation dataset.

3. Results

3.1. Offline validation results

Table 1 shows the offline performance of different models we have tried based on our baseline model. Model 1 is our baseline model that uses SE_ResNet as the framework. In Model 2, we apply wavelet denoising, add the relabeled CPSC data to training data, and a rule-based model for bradycardia. The performance of Model 2 improved in both our offline validation dataset and Hefei validation set, compared to our baseline model. However, we found that the problem of threshold shifting remained. To stabilize the threshold and enhance the generalizability of our model, we introduce Sign Loss to Model 3 and apply NSR post-processing. Though Model 3 shows an inferior performance on our offline validation dataset, it shows better performance on the Hefei validation set. To some extent, it can be explained that Sign Loss can improve the model's generalization ability. Based on Model 3, Model 4 and Model 5 only use eight leads signal data in 12-lead ECG to improve model training efficiency. The length of signal input for Model 4 and Model 5 is 15000 and 5000, respectively. Both models' training time is less than that of Model 3, and the performance remains unchanged. Considering the training efficiency, model performance, and model generalization, we ensemble Model 4 and Model 5 to form Model 6. Model 6 obtain a score of 0.683 on the offline validation dataset and a score of 0.319 on the Hefei validation set, which is better than both Model 4 and Model 5. Therefore, we select Model 6 as our best model.

3.2. Online testing results

Table 2 shows the official evaluated challenge scores and rank on (a) official validation dataset, (b) official test database 1 and (c) official test database 2, (d) official test database 3 and (e) the entire official test dataset. The official challenge ranking demonstrated the model's ability to classify the ECG abnormalities despite the challenges presented, e.g. noise in the signals and labels. Meanwhile, the difference between the official validation score and the offline validation score is only 0.001, suggesting good generalizability and little overfitting of the proposed model.

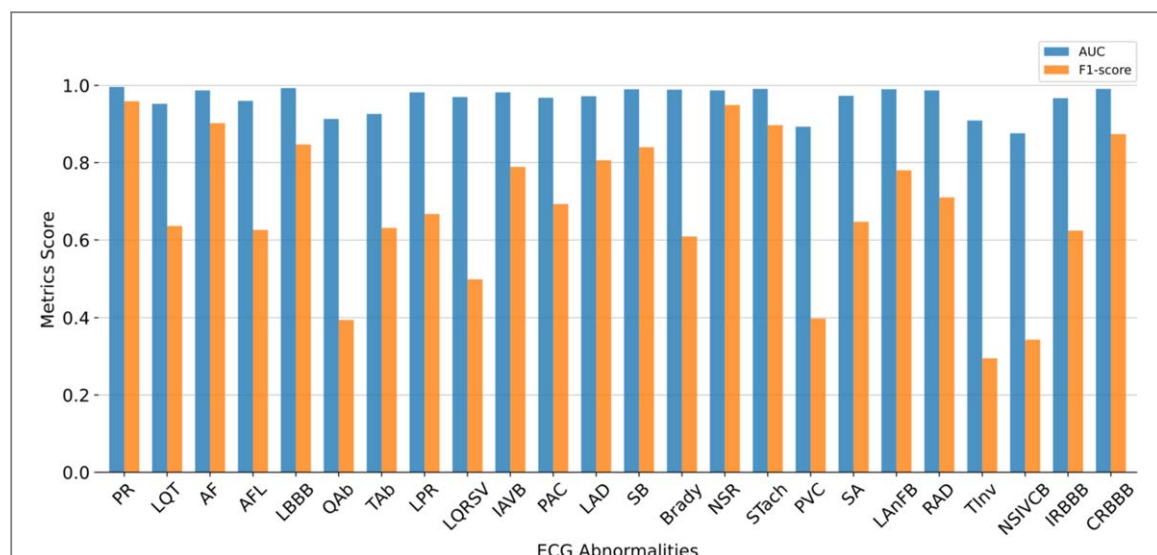


Figure 4. The best model's performance on each ECG abnormality (full name shown in appendix table A2) in area under curve (AUC) and F1-score. The AUC measures the model's ability to identify the positive and negative samples when considering each ECG abnormality. The F1-score evaluates the model's performance of multi-label classification, which considers all abnormalities. From the figure, we can see that the AUC for each ECG abnormality is relatively high, indicating that our model can classify each ECG abnormality well when there is no intervention from other abnormalities. The fluctuation of the F1-score shows performance decreases when all abnormalities are considered. This could be due to the high similarity of different ECG abnormalities feature spaces that tend to affect the model.

Table 2. Performance of our best model on official evaluation datasets.

Official evaluation datasets	Score	Official challenge ranking
Official validation dataset	0.682	3
Official test database 1	0.852	3
Official test database 2	0.649	2
Official test database 3	0.396	3
The entire official test dataset	0.514	3

3.3. Detailed model performance analysis

There are 27 ECG abnormalities in the official evaluation, of which three pairs were treated as the same ECG abnormality when the challenge organizers calculating the score. These three pairs are complete right bundle branch block and right bundle branch block, premature atrial contraction and supraventricular premature beats, premature ventricular contradictions, and ventricular premature beats (VPB). Based on this design, we analyzed our models performance in 24 categories. Figure 4 shows the performance of our proposed method on each ECG abnormality, from which we found the factors that may affect the models performance, shown below.

- (1) *Partial label*: The AUC of each ECG abnormality is generally high, while the F1-score of some ECG abnormality is at a low level. Likely, some anomalies in the data are not labeled, which leads to an excessive prediction of false positives. Here are two possible reasons. Firstly, there are six datasets in total, each dataset has only partial abnormalities, and no dataset has all 27 abnormalities. For example, atrial fibrillation and sinus rhythm appear in all six datasets, with complete annotation and good overall model performance. However, premature ventricular contractions and low QRS voltages only appear in two datasets, thus the models performance is relatively low. The second reason could be annotation error, where some abnormalities in a dataset are missing-labeled or wrongly labeled.
- (2) *Hard-to-detect features*: Some characteristics of ECG abnormalities are hard to detect. For example, for some cases in low QRS voltages, we found that the amplitude of the signal differs greatly. This could be due to the difference in weights and heights habitus of patients, which affects the resistivity when sampling the ECG signals.

- (3) *Feature confusion*: We also found that characteristics between two ECG abnormalities could be too similar for the model to classify. For instance, the characteristic of bradycardia is similar to sinus bradycardia since both of them show a slow heart rate.

As we mentioned previously, the SE module in SE_ResNet can obtain the importance of each feature by learning, and then enhance the useful features according to the weightage and suppress the features that are not useful for the current task, so the performance of the model can be improved compared with the original ResNet. In our proposed method, we integrate two SE_ResNet, which takes eight-lead ECG signal data with a length of 5000 and 15000 as input, respectively. The integrated model outperforms the two sub-models, which shows that the integrated model effectively combines the advantages of the two different input length settings. Firstly, the length of the ECG samples in the datasets ranges from 2500 to over 100000. The larger input length can contain more information in samples with longer signal length, while the smaller input length can make samples with short length free from the information losses of padding in training. Secondly, some of the ECG abnormalities show a characteristic of continuous repeat, while some occasionally appear. Hence, the larger input length can capture the intermittent abnormal signal, while the smaller input can reduce the difficulty of the neural network for anomaly detection.

4. Discussion

Existing deep learning methods for ECG classification mainly fall into two categories: CNN-based method and sequence model-based method.

Compared to other CNN-based methods Min *et al* (2020), Chen *et al* (2020), Jia *et al* (2020) in the challenge, our proposed ensemble model has several advantages, including

- (1) A series of pre-processing methods to mitigate the problem of data divergence across various sources.
- (2) SE modules that capture channel-wise relations thus improve the classification accuracy.
- (3) Model fusion mechanism that effectively improves the model's generalizability.
- (4) Sign Loss function that tackles the class imbalance thus improve the model's robustness.

However, an intrinsic property of CNN is that it focuses on morphological features and likely to lose temporal information to some degree. Sequence models, on the other hand, are built to extract temporal patterns. Some other teams in the challenge did employ the transformer Vaswani *et al* (2017) structure to capture longer temporal dependencies of the ECG signal, and they have achieved promising results as well. Therefore, we plan to investigate along this direction further in our future work.

In addition, the top-2 solutions Natarajan *et al* (2020), Zhao *et al* (2020) also take the static feature such as age and gender into account, while we did not. Such demographics information may help the model to learn some patient-specific ECG patterns.

Inspired by some of the promising ideas from the other top teams in the challenge, we, as future work, plan to improve our work in the following aspects: (1) to combine ECG medical knowledge with deep learning to construct a classification model. For example, use machine learning methods instead of traditional methods that are not accurate enough to extract medical features such as R-R intervals, and combine them with the features extracted by SE_ResNet to improve the interpretability and generalizability. (2) To develop a better model for locating the abnormal signals. Many ECG abnormalities occur intermittently, and it is a challenging task to locate the timing of abnormal signals in a long period of ECG data. An accurate locating of abnormalities present in lengthy ECG data could help to better diagnosis in clinical practice.

5. Conclusions

In this paper, we propose a deep learning framework to automatically identify multiple ECG abnormalities. Compared to previous works, the main contribution of our methods is three-fold. Firstly, our proposed framework can classify 27 types of ECG abnormalities on eight-lead ECG signals, while previous works focused on at most nine types of ECG abnormalities classification using single-lead ECG signals. Secondly, we introduce a Sign Loss to mitigate the class imbalance problem and improve our framework's generalizability. Thirdly, our framework is developed on six different datasets across various countries, and we proposed several pre-processing methods to target the diversity issue from different data sources, whereas previous works mainly use a

small dataset from a single data source. Our proposed framework is developed and validated on real-world datasets, and we believe it has the potential to be deployed in clinical practice.

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Appendix

Table A1. Basic information of six different datasets that forms the challenge training dataset.

Databases	ECG recordings	ECG abnormality categories
CPSC	6,877	9
CPSC2	3453	72
INCART	74	37
PTB	516	17
PTB-XL	21 837	50
Georgia	10 344	67

Table A2. 27 ECG abnormalities and their corresponding abbreviations.

ECG abnormality	Abbreviation
1st degree AV block	IAVB
Atrial fibrillation	AF
Atrial flutter	AFL
Bradycardia	Brady
Complete right bundle branch block	CRBBB
Incomplete right bundle branch block	IRBBB
Left anterior fascicular block	LAnFB
Left axis deviation	LAD
Left bundle branch block	LBBS
Low QRS voltages	LQRSV
Nonspecific intraventricular conduction disorder	NSIVCB
Pacing rhythm	PR
Premature atrial contraction	PAC
Premature ventricular contractions	PVC
Prolonged PR interval	LPR
Prolonged QT interval	LQT
Q wave abnormal	QAb
Right axis deviation	RAD
Right bundle branch block	RBBB
Sinus arrhythmia	SA
Sinus bradycardia	SB
Sinus rhythm	NSR
Sinus tachycardia	STach
Supraventricular premature beats	SVPB
T wave abnormal	TAB
T wave inversion	TInv
Ventricular premature beats	VPB

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